# VEGETATIVE MORPHOLOGY & ANATOMY

*Detailed Notes — Roots, Stems & Leaves*

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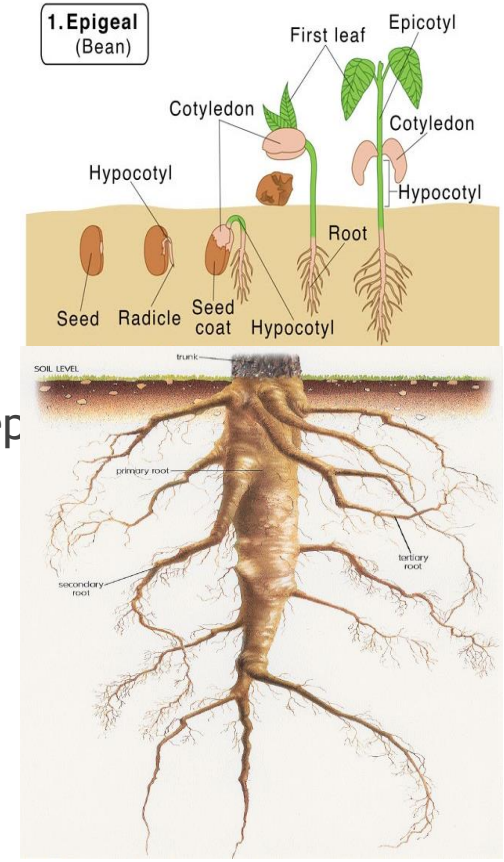
# 1. Roots

- A root is the part of the plant axis that develops from the radicle of the embryo and normally grows downward into the soil. Unlike stems, roots lack nodes and internodes, never bear leaves or buds directly on their surface, are usually non-green, and are **positively geotropic (grow toward gravity)** and **positively hydrotropic (grow toward water)**, while being **negatively phototropic (grow away from light)**.
- **Functions:** anchorage of the plant in soil; absorption of water and dissolved mineral salts; conduction of water/minerals up to the stem; storage of food (in many species); and, in some plants, vegetative propagation.



# 1.1 Root Types: Taproot vs. Fibrous Systems

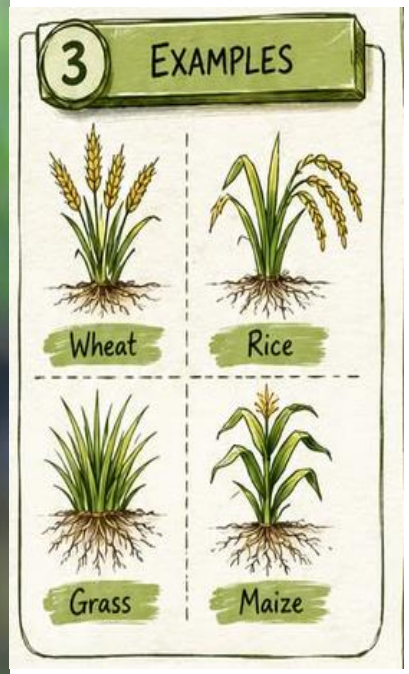
- ***Taproot system***
- Develops directly from the radicle, which persists and continues to grow vertically as the dominant, thick **primary (main) root**.
- The primary root gives off thinner **secondary** and **tertiary lateral roots**, producing a branched, tree-like system that penetrates deep into the soil.
- Typical of most **dicotyledonous** plants — e.g., mango, mustard, gram, carrot.
- Because it reaches deep soil layers, it anchors the plant firmly and can access deep water reserves; it is also frequently modified for food storage (see §1.4).



# 1.1 Root Types: Taproot vs. Fibrous Systems

- ***Fibrous root system***
- The short-lived radicle is soon replaced by a cluster of thin, moderately branched **adventitious roots** that arise from the base of the stem.
- All the roots are of roughly similar thickness and length — there is no single dominant root.
- Typical of most **monocotyledonous** plants — e.g., maize, rice, wheat, and other grasses.
- Spreads widely through the shallow topsoil, making it very efficient at absorbing surface water and nutrients and at binding topsoil (important for controlling soil erosion — relevant to agricultural soil conservation).

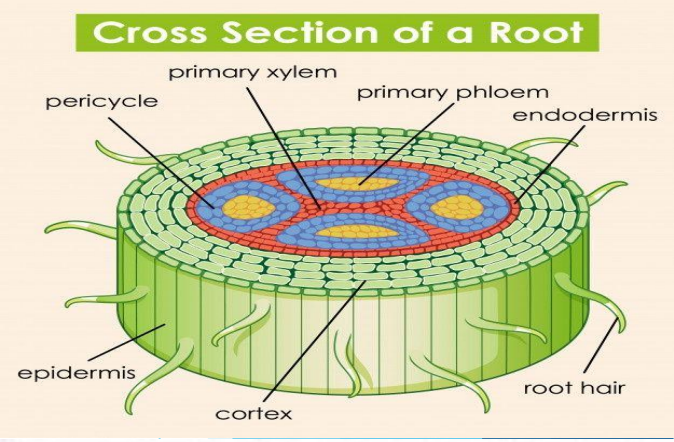


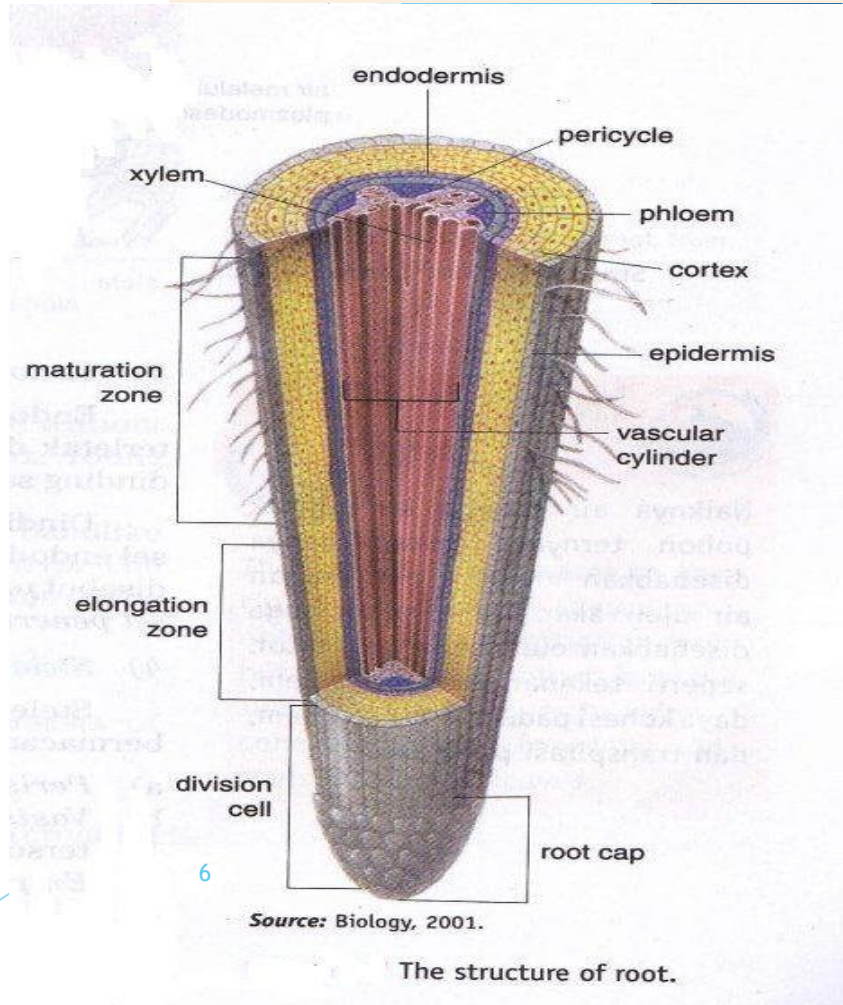


# 1.2 Primary and Secondary Root Internal Structures

- ***Primary structure (transverse section of a young root)***

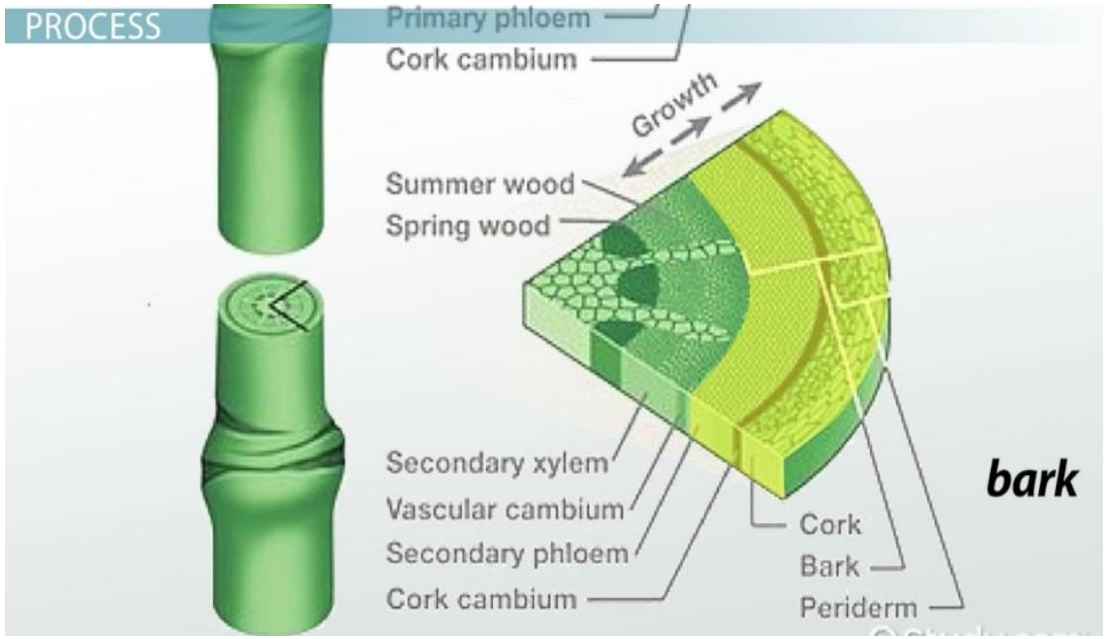
- **Epidermis (piliferous layer):** the single outermost layer; thin-walled, without a cuticle over the absorptive region so that water can pass through easily; produces tubular outgrowths called **root hairs** that greatly increase the absorptive surface area.
- **Cortex:** several layers of thin-walled, loosely packed **parenchyma** cells with intercellular air spaces; stores food and allows lateral movement of water toward the centre.
- **Endodermis:** the innermost, single-layered boundary of the cortex; its radial and transverse walls carry a band of suberin/lignin called the **Casparian strip**, which forces water and dissolved minerals to pass through the cell membranes (not between cells) — giving the plant control over what enters the vascular tissue.
- **Pericycle:** a layer (or few layers) of thin-walled cells just inside the endodermis; it is meristematic and is the tissue from which **lateral roots** originate, and later, part of the vascular cambium and cork cambium during secondary growth.
- **Vascular tissue (stele):** primary **xylem** forms a solid, star-shaped mass at the very centre, with the tips of the star (protoxylem) pointing outward toward the pericycle and the wider metaxylem toward the middle (an arrangement called exarch); primary **phloem** occurs as separate patches lying between the arms of the xylem star. Roots are classed as diarch, triarch, tetrarch, etc., according to the number of xylem arms.



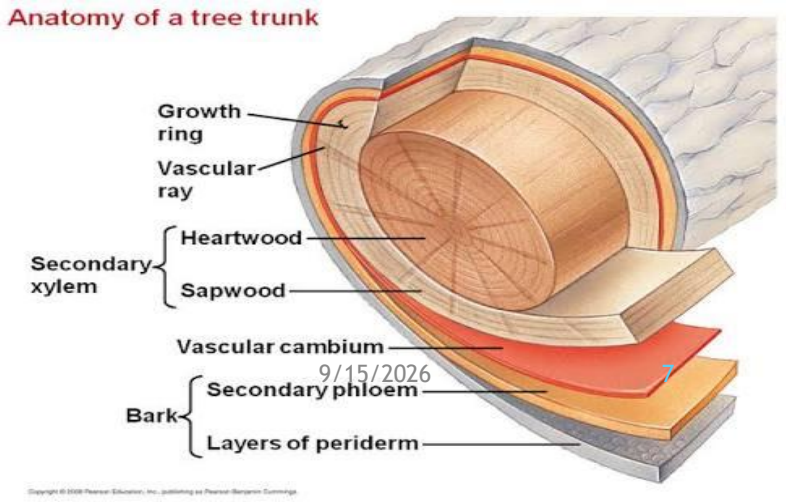

The structure of root.

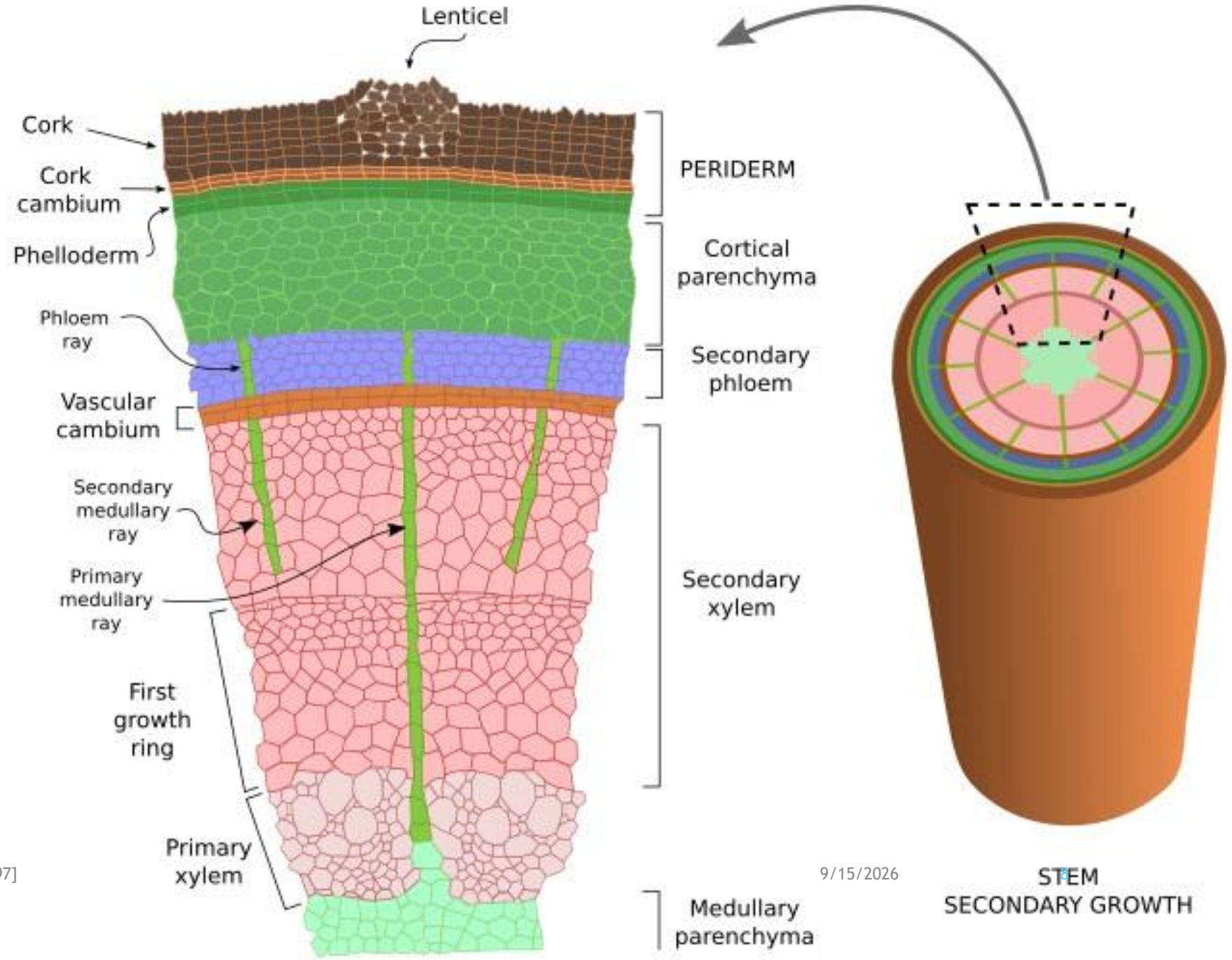
# 1.2 Primary and Secondary Root Internal Structures




A **cork cambium (phellogen)** arises in the pericycle; it cuts off **cork (phellem)** to the outside and **phelloderm** to the inside, together forming a protective periderm that eventually replaces the epidermis and cortex.
This secondary thickening is responsible for the **girth increase** seen in woody roots, and for the concentric growth of many storage roots (e.g., carrot, sweet potato).

- ***Secondary structure (secondary growth)***
- Occurs in roots and stems of dicots (and gymnosperms) that undergo thickening; monocot roots normally lack secondary growth.
- A **vascular cambium** develops (from pericycle and tissue between the xylem and phloem) and produces **secondary xylem (wood)** toward the inside and **secondary phloem** toward the outside, season after season.

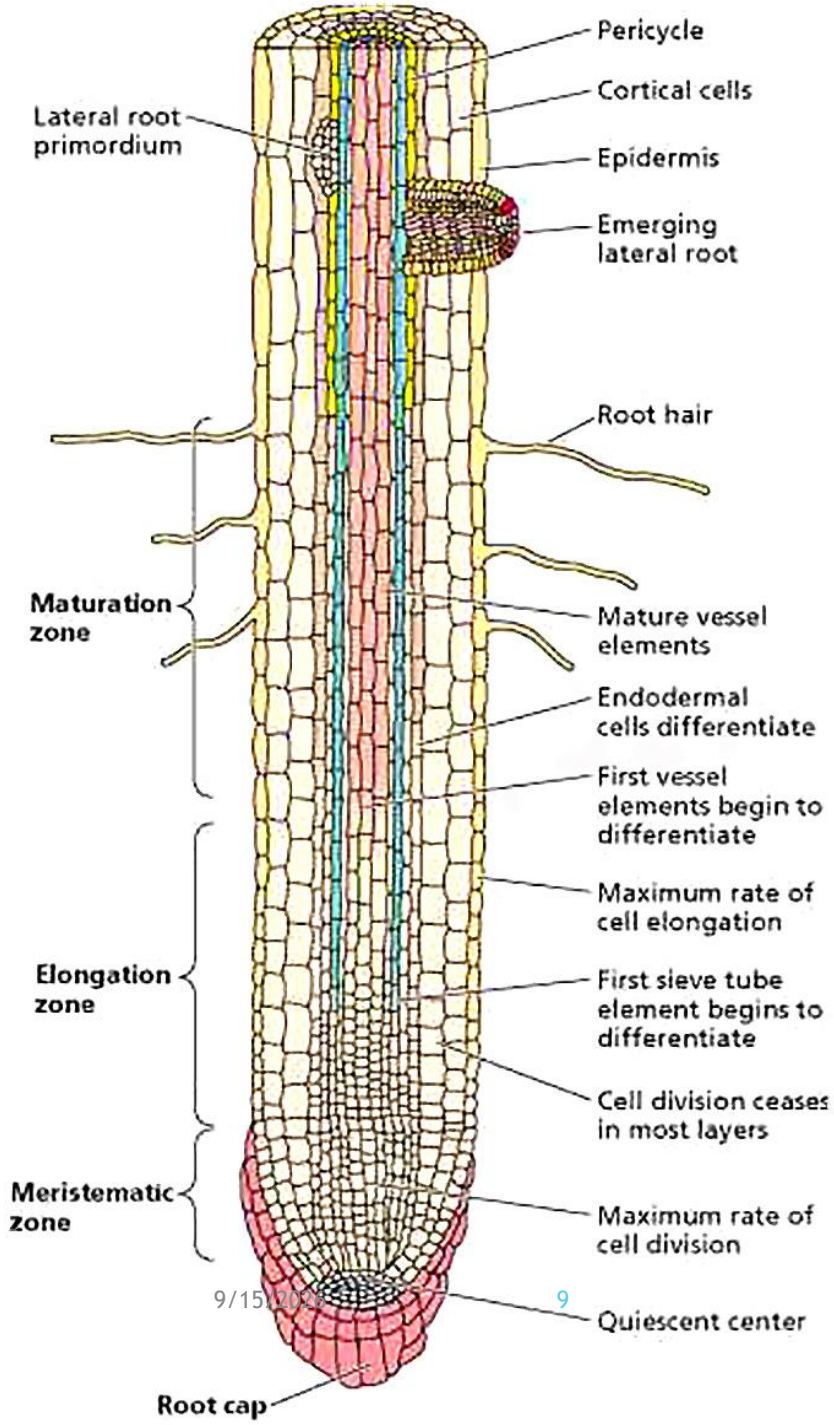



Lenticel
Cork
Cork cambium
Phelloderm
Phloem ray
Vascular cambium
Secondary medullary ray
Primary medullary ray
First growth ring
Primary xylem
PERIDERM
Cortical parenchyma
Secondary phloem
Secondary xylem
Medullary parenchyma
STEM
SECONDARY GROWTH

# 1.3 Root Zone Regions

Viewed in longitudinal section, the growing tip of every root shows four zones that grade gradually into one another:

- **Root cap**: a thimble-shaped mass of parenchyma cells covering and protecting the delicate apical meristem as it is pushed through abrasive soil particles; outer cells are constantly worn away and replaced from within; it secretes a slimy **mucilage** that lubricates the root's passage, and its inner cells (statocytes) contain starch grains that help the root sense and respond to gravity.
- **Zone of cell division (meristematic zone):** lies just behind the root cap; consists of small, thin-walled, densely cytoplasmic cells with large nuclei that divide rapidly by mitosis, continuously adding new cells to both the root cap and the zone above.
- **Zone of elongation**: cells produced by the meristem stop dividing and elongate rapidly (mainly through vacuolation), and it is this elongation that provides the main force pushing the root tip forward through the soil.
- **Zone of maturation (zone of differentiation / root hair zone):** cells stop elongating and differentiate into the permanent tissues described in §1.2; epidermal cells here produce **root hairs**, and the primary xylem and phloem become functional, allowing the region to begin absorbing and conducting water.



# 1.4 Root Modifications

- ***Storage roots***

- Roots (usually taproots) become swollen with stored food (mainly starch and sugars), typically forming shapes such as: **fusiform** — spindle-shaped, broad in the middle and tapering at both ends (e.g., radish); **napiform** — almost spherical, broad at the top and abruptly tapering below (e.g., turnip, beetroot); **conical** — broad at the top, gradually tapering to a point (e.g., carrot).



- **Tuberous roots** are adventitious roots that swell irregularly (not in a smooth taper) — e.g., sweet potato, cassava, dahlia.





# 1.4 Root Modifications



banyan fig

***Roots modified for support***

**Prop (stilt) roots**: thick adventitious roots that arise from the lower nodes of the stem and grow obliquely into the soil, propping up tall stems that would otherwise be top-heavy — e.g., maize, sugarcane, and the massive pillar-like prop roots of the banyan fig.

**Buttress roots**: large, plank-like, above-ground roots that radiate from the base of the trunk in many big tropical trees, giving extra stability in shallow tropical soils.



# *Other modifications (brief)*

- **Climbing roots**: short adventitious roots that cling to a support surface, e.g., betel, money plant (Pothos).

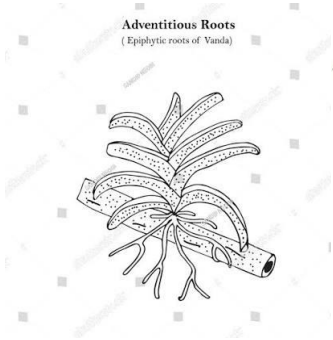


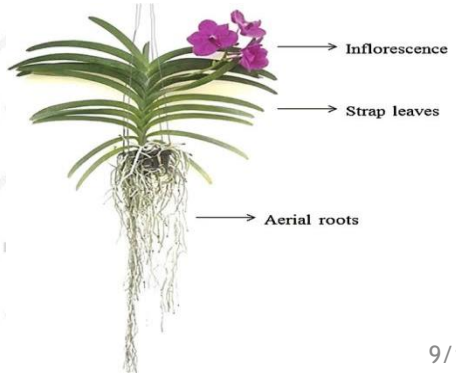



- **Respiratory roots (pneumatophores):** grow upward out of waterlogged, oxygen-poor mud into the air for gas exchange, e.g., mangroves.



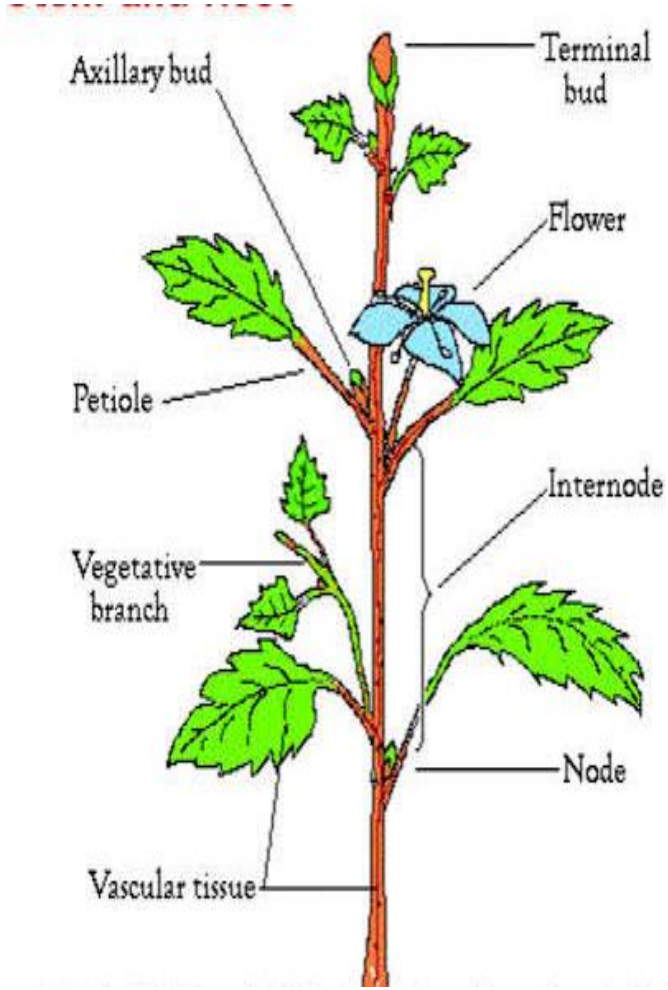
- **Epiphytic (velamen) roots**: spongy-coated aerial roots that absorb atmospheric moisture, e.g., orchids.



# 2. Stems

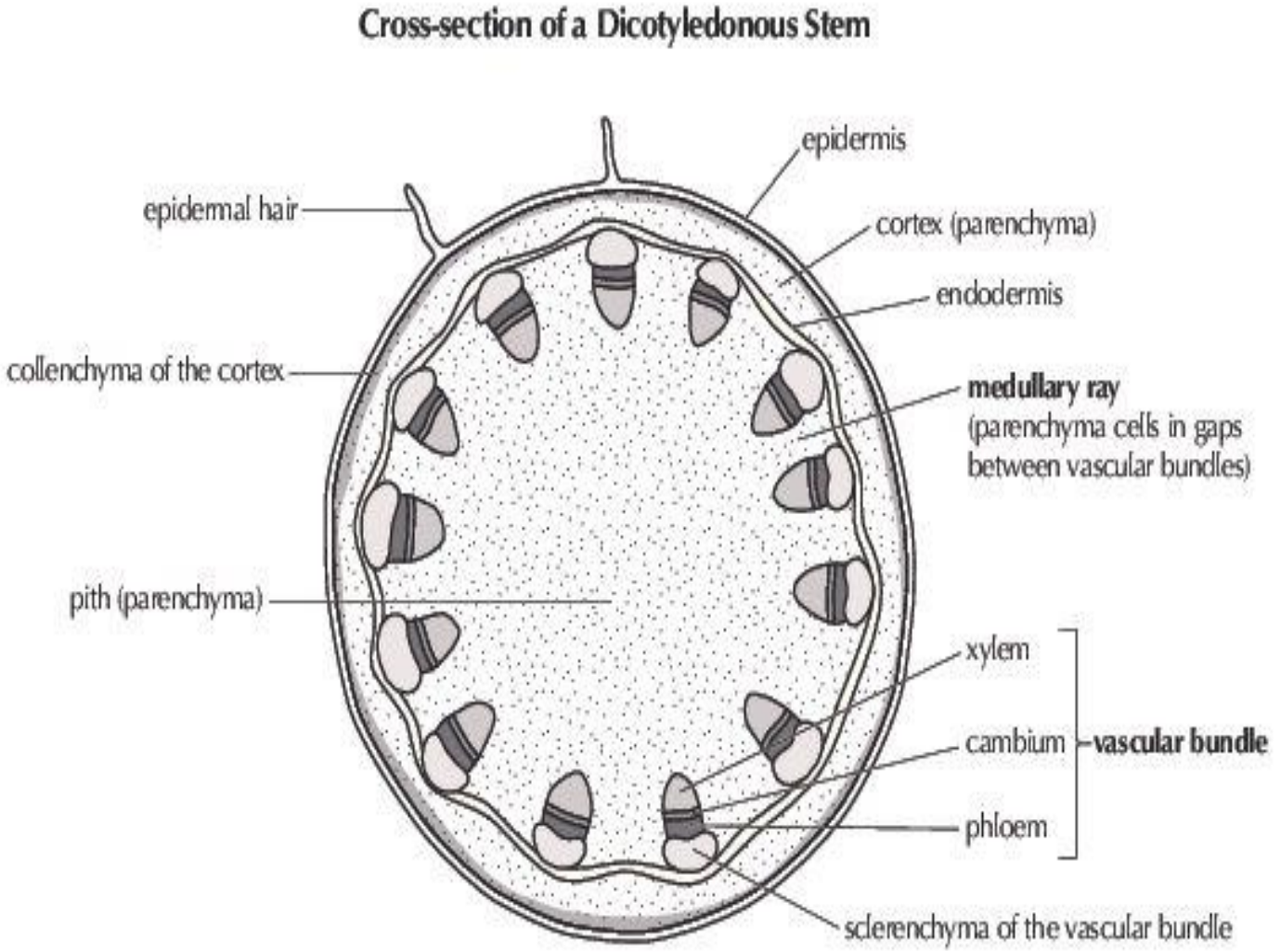
- The stem is the part of the plant axis that develops from the plumule of the embryo and usually grows upward, bearing leaves, buds, flowers, and fruits. Its surface is marked by distinct **nodes** (points where leaves/buds are attached) and **internodes** (the stretches of stem between successive nodes) — a feature that immediately distinguishes it from a root.
- **Functions:** mechanical support and display of leaves for maximum light capture; conduction of water and minerals upward (via xylem) and manufactured food downward/throughout the plant (via phloem); bearing reproductive structures; and, in many species, storage or vegetative propagation.



# 2.1 Dicot vs. Monocot Stem Anatomy

***Dicot stem (herbaceous, transverse section)***

- **Epidermis:** single outer layer, usually covered by a cuticle, with occasional stomata and hairs (trichomes).
- **Cortex:** differentiated into an outer **hypodermis** (collenchyma, providing mechanical strength while allowing growth), a middle region of parenchyma (sometimes chlorenchyma, capable of photosynthesis), and an inner boundary layer sometimes called the **endodermis / starch sheath**.
- **Pericycle:** often includes patches of sclerenchyma (called "hard bast") that add mechanical strength just outside the vascular bundles.
- **Vascular bundles:** arranged in a single, well-defined **ring** around a large central pith; each bundle is **conjoint** (xylem and phloem in the same bundle), **collateral** (phloem lies to the outside of the xylem), and **open**, meaning a strip of **cambium** lies between the xylem and phloem — this cambium is what allows dicot stems to undergo secondary (girth) growth.
- **Pith:** a large central zone of parenchyma occupying the centre of the stem, often used for storage; narrow **medullary (pith) rays** of parenchyma run between adjacent vascular bundles, connecting the pith to the cortex.



# 2.1 Dicot vs. Monocot Stem Anatomy

***Monocot stem (transverse section)***

- **Epidermis**: covered by a relatively thick cuticle; a mechanically strong, sclerenchymatous **hypodermis** lies just beneath it (rather than the collenchyma typical of dicots).
- **Ground tissue**: forms one continuous mass of parenchyma with no clear boundary between "cortex" and "pith" — the two are not differentiated as they are in dicots.
- **Vascular bundles**: numerous and **scattered** irregularly throughout the ground tissue (more crowded toward the periphery), rather than arranged in a ring.
- Each bundle is conjoint, collateral, and **closed** — there is **no cambium**, so monocot stems normally cannot thicken through secondary growth. Each bundle is typically surrounded by a tough **sclerenchymatous bundle sheath**, and its xylem vessels are often arranged in a Y- or V-shape with a conspicuous protoxylem lacuna (cavity) at the centre of the V.

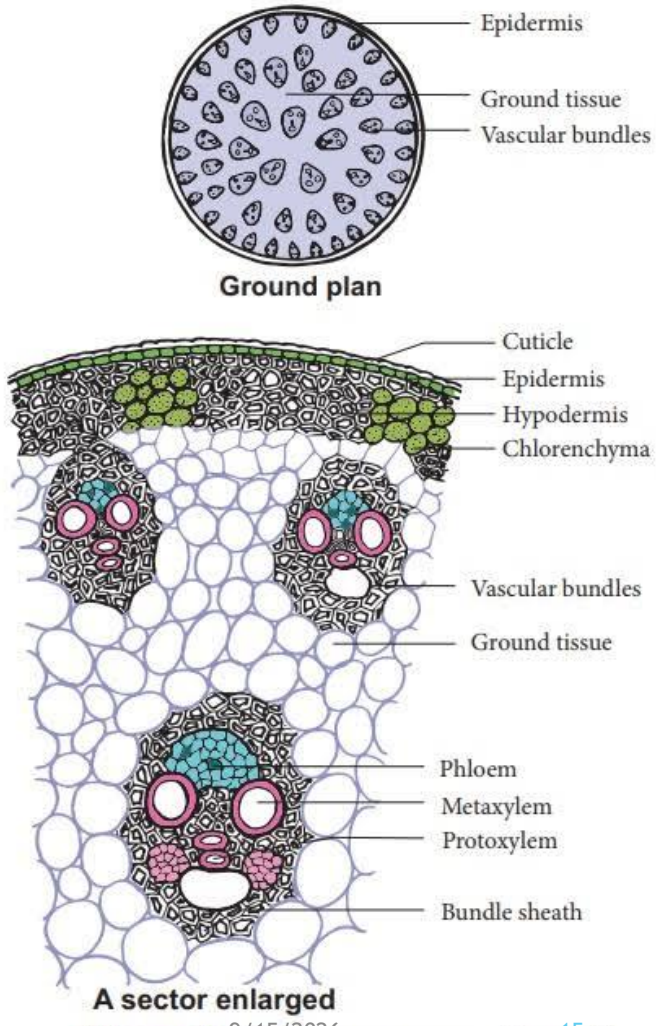



Figure 12.5 Transverse section of Monocot stem

# 2.2 Vascular Bundle Arrangements — Key Terms

- **Conjoint bundle:** xylem and phloem occur together within the same vascular bundle (typical of stems and roots' aerial parts) — contrast with **radial bundles**, in which xylem and phloem alternate around the stele rather than being bundled together, as seen in roots.
- **Collateral bundle:** phloem lies on one side of the xylem only (usually the outer side) — the common arrangement in both dicot and monocot stems.
- **Bicollateral bundle:** phloem occurs on both the outer and inner sides of the xylem, e.g., in the Cucurbitaceae (pumpkin/cucumber family) and Solanaceae (e.g., potato, tomato).
- **Open bundle:** cambium present between xylem and phloem, permitting secondary growth (dicot stems).
- **Closed bundle:** no cambium; no secondary growth possible (monocot stems).
- **Overall arrangement in the stem:** ring arrangement (dicot stems) vs. scattered arrangement (monocot stems); compare this with the radial arrangement of xylem and phloem found in roots (§1.2).

# 2.3 Stem Modifications

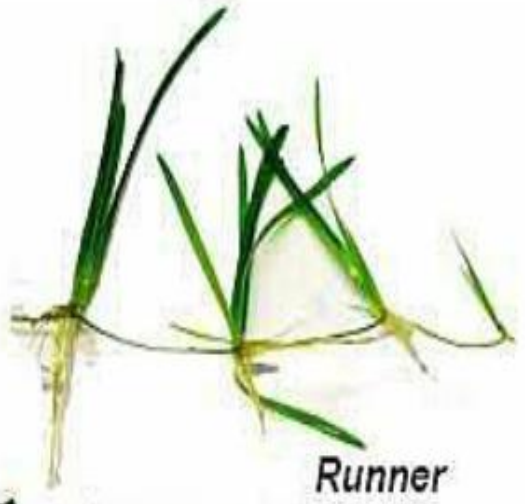
- ***Underground stem modifications***

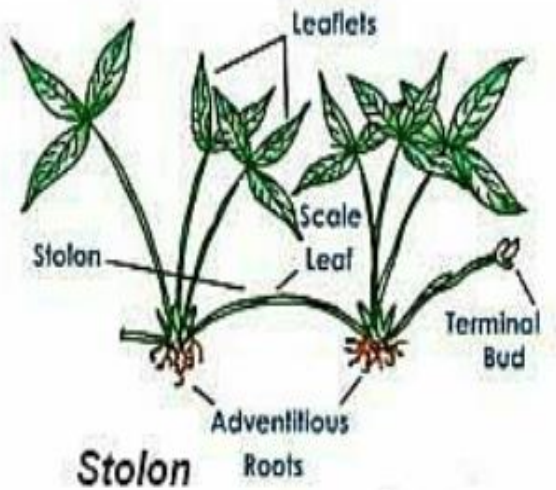
- **Rhizome**: a horizontal, usually thickened underground stem that grows just below the soil surface; it shows distinct nodes and internodes, bears small scale leaves and axillary buds at the nodes, and produces adventitious roots from its lower surface — e.g., ginger, turmeric, banana "corm" structures.
- **Tuber**: the swollen tip of an underground stolon, packed with stored starch; its surface bears small depressions called "**eyes**," which are actually nodes, each with a dormant **axillary bud** capable of sprouting a new shoot — e.g., the potato (a classic example used to teach that a tuber is a stem, not a root, precisely because it bears buds/eyes).
- Related (not explicitly listed but useful to know): **corm** — a short, vertical, disc-like underground stem (e.g., taro/cocoyam), and **bulb** — a short stem ("basal plate") surrounded by fleshy, food-storing scale leaves (e.g., onion, garlic).

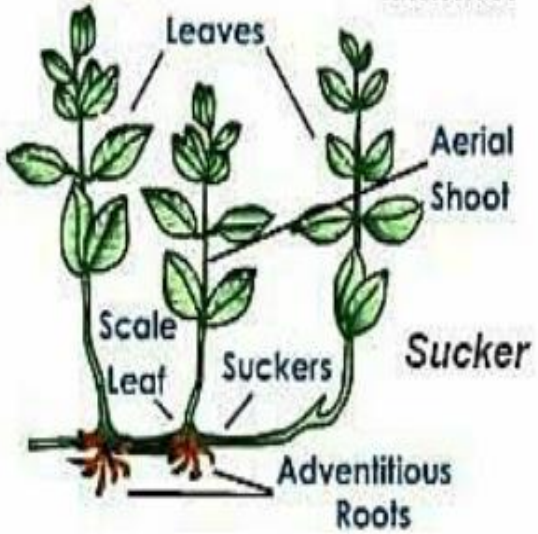


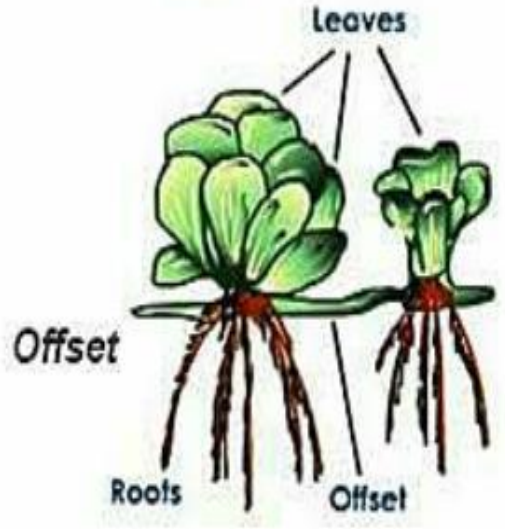
# 2.3 Stem Modifications

- ***Subaerial / surface stem modifications***
- **Stolon (runner):** a slender stem that grows horizontally along (or just above) the soil surface, with long internodes; at each node it can put down adventitious roots and produce a new, independent plantlet — e.g., strawberry, couch grass, mint.
- **Sucker**: similar to a rhizome but grows obliquely, arising from the base of the main stem and emerging above ground some distance away to form a new shoot — e.g., banana, chrysanthemum, pineapple.
- **Offset**: a short, thick, horizontal branch found mainly in aquatic plants, bearing a rosette of leaves and roots at its tip — e.g., water hyacinth, water lettuce (Pistia).



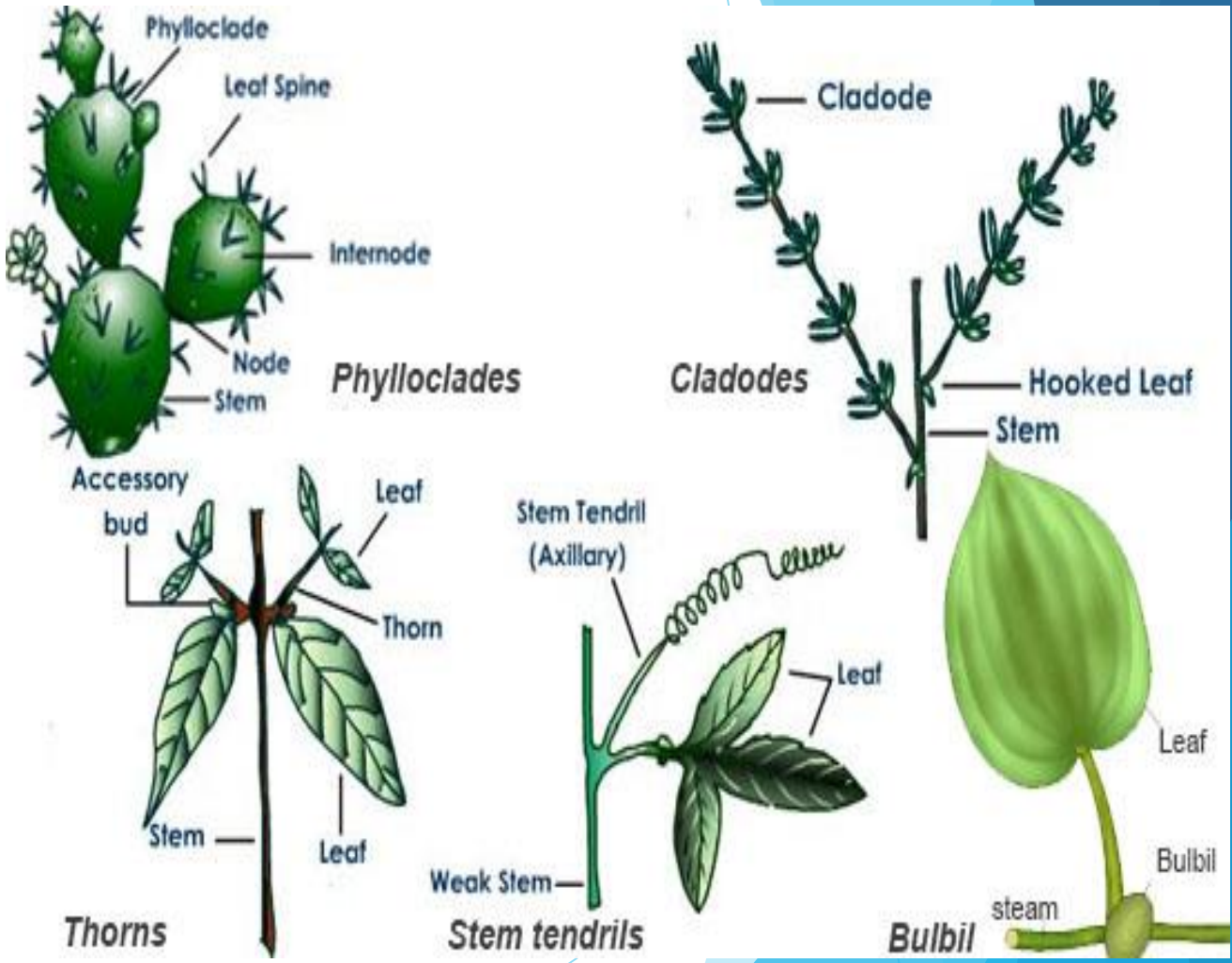



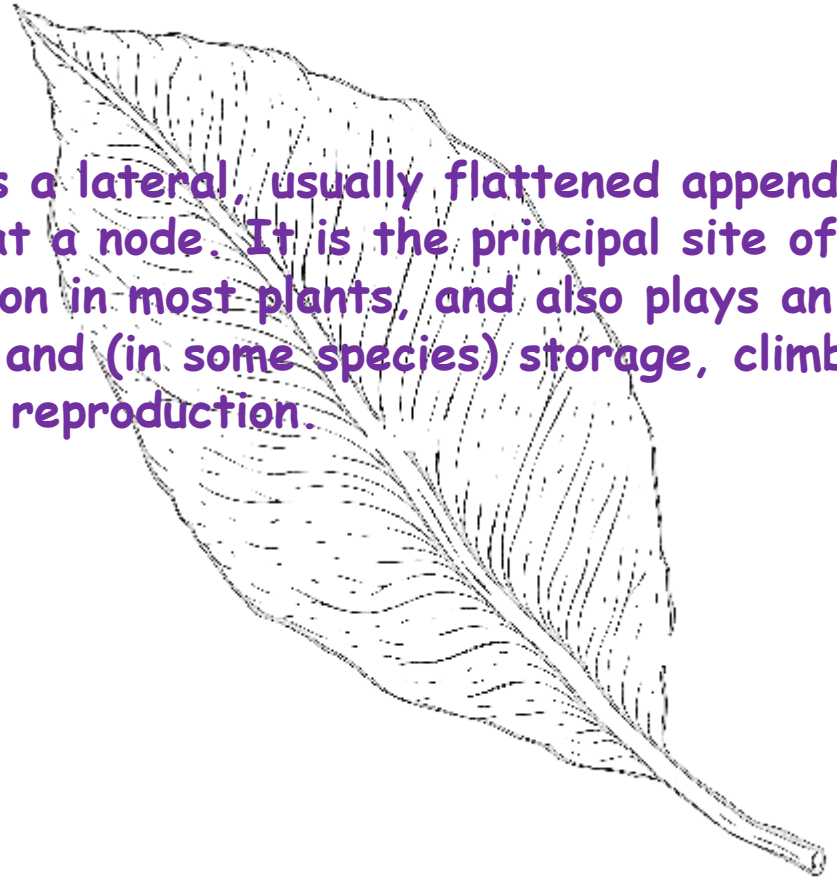



# Other aerial modifications (brief)

- **Stem tendrils:** slender, coiling structures used for climbing, e.g., grapevine, passion fruit.
- **Thorns:** hard, pointed structures for defence against browsing animals, e.g., bougainvillea, citrus.
- **Phylloclades / cladodes:** flattened, green, photosynthetic stems that take over the leaves' function in dry habitats, with the true leaves reduced to spines — e.g., Opuntia (prickly pear cactus), Asparagus.



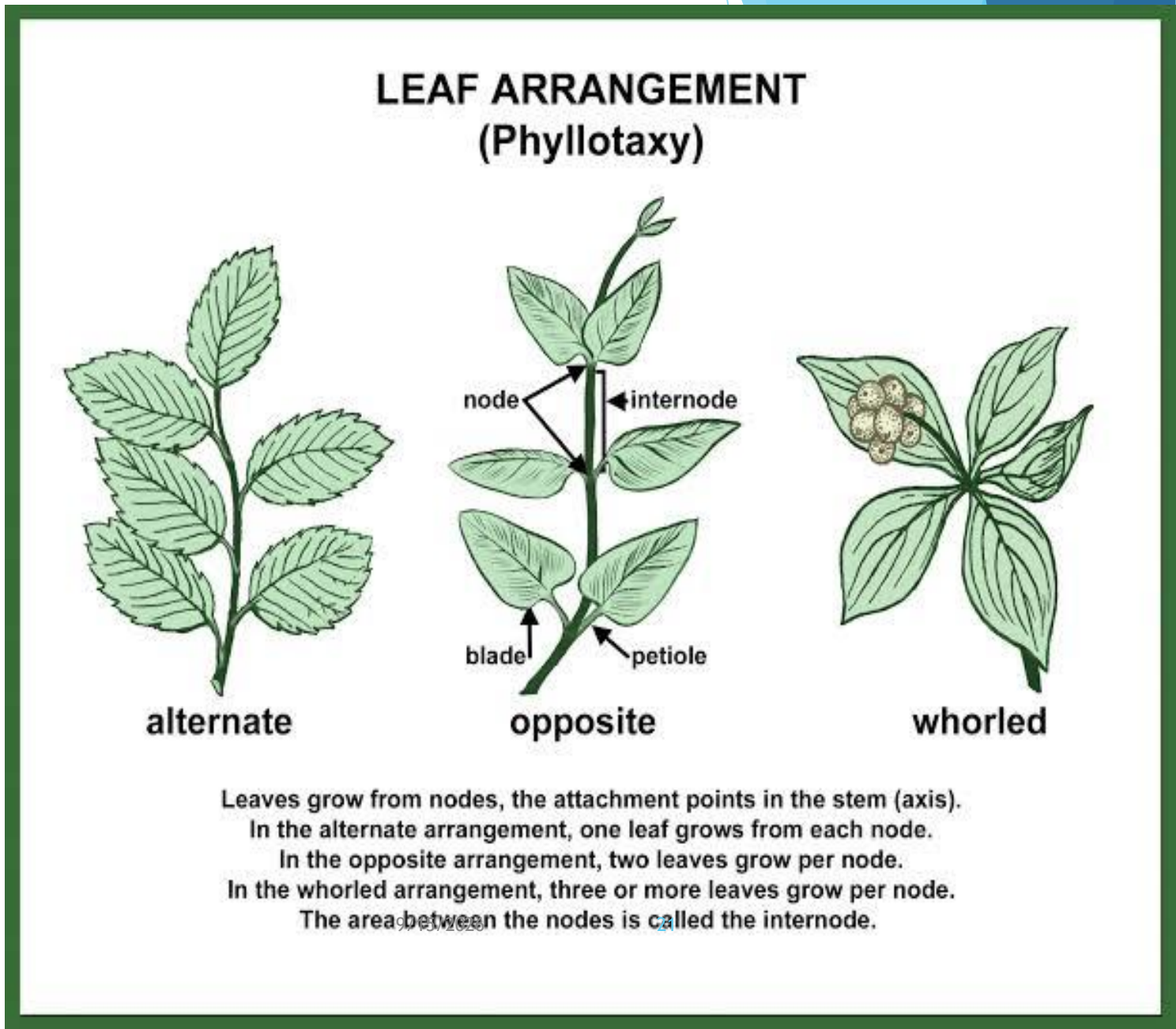
# 3. Leaves



- The leaf is a lateral, usually flattened appendage that develops from the stem at a node. It is the principal site of photosynthesis and transpiration in most plants, and also plays an important role in gas exchange, and (in some species) storage, climbing support, defence, or vegetative reproduction.

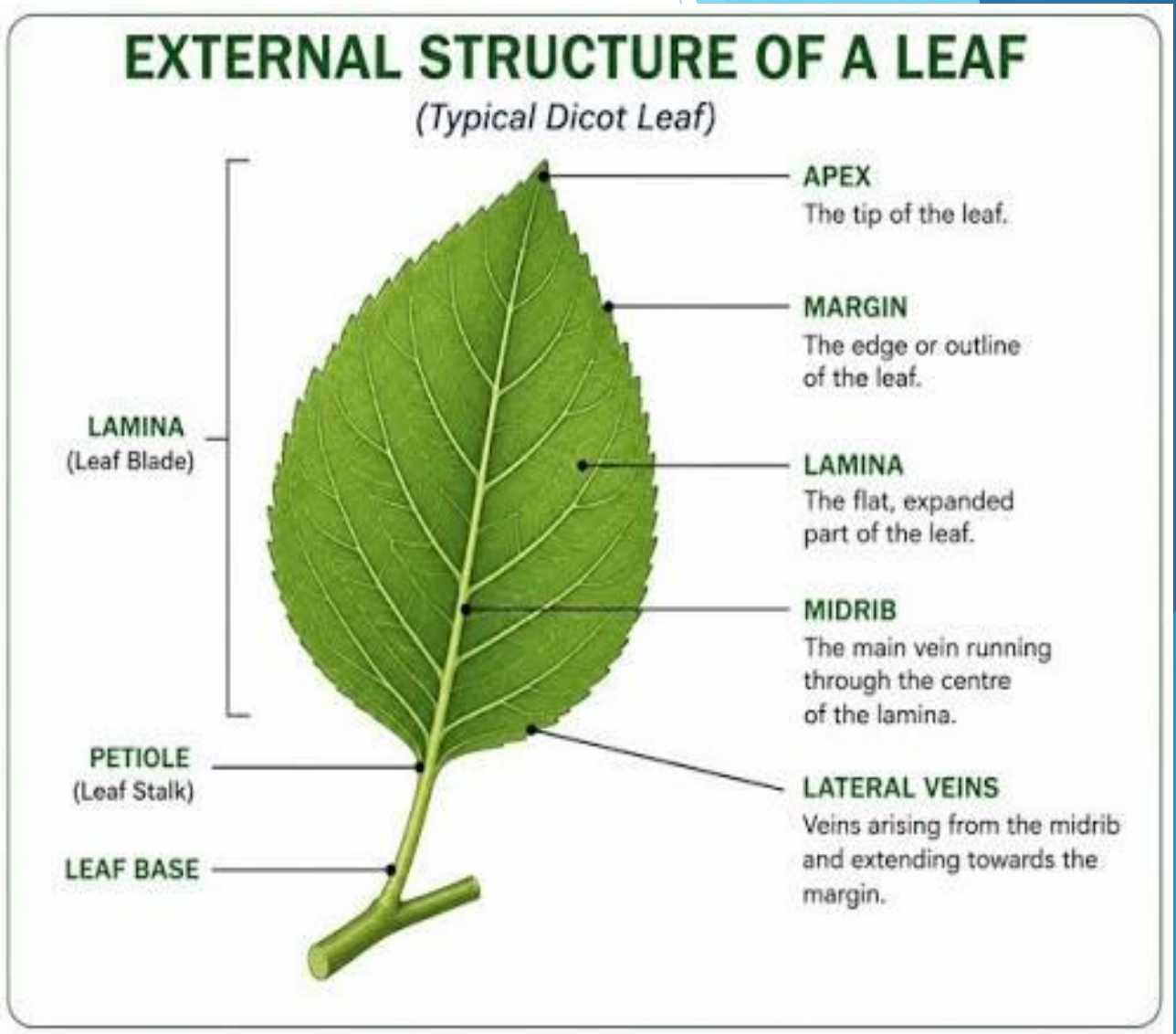
# 3.1 Leaf Arrangement (Phyllotaxy)

- Phyllotaxy is the pattern in which leaves are arranged on the stem. Because it determines how much each leaf shades the ones below or beside it, phyllotaxy has a direct effect on how efficiently the plant captures sunlight.
- **Alternate (spiral):** a single leaf is borne at each node, with successive leaves arising on alternating sides of the stem, tracing a spiral path up the stem when viewed from above — the commonest pattern, e.g., hibiscus, sunflower, mustard.
- **Opposite:** two leaves are borne at each node, directly facing each other on opposite sides of the stem. Where successive pairs are rotated 90° to the pair below (**decussate** arrangement), shading of lower leaves is minimised — e.g., Calotropis, guava, and most members of the mint family.
- **Whorled:** three or more leaves arise together from a single node, forming a ring around the stem — e.g., Alstonia, oleander (Nerium).



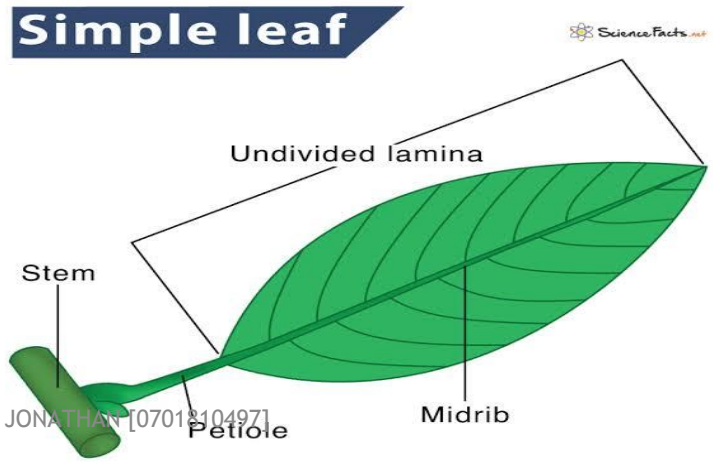
# 3.2 External Structures

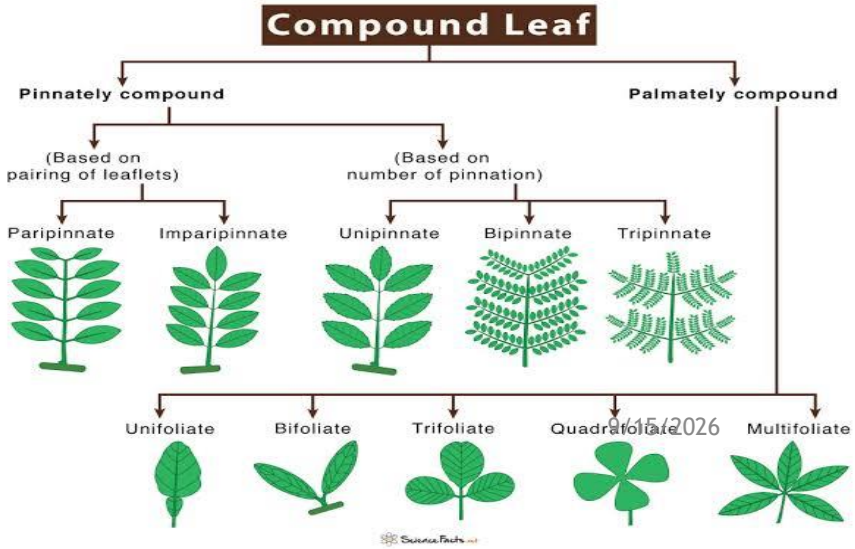
- **Leaf base:** the point where the leaf attaches to the stem; may bear a pair of small leaf-like outgrowths called **stipules**, or expand into a sheath that wraps around the stem (typical of grasses).
- **Petiole (leaf stalk):** connects the base to the blade and orients the blade toward light; leaves with a petiole are **petiolate**, while those attached directly to the stem without one are **sessile**.
- **Lamina (blade):** the flattened, expanded, usually green part of the leaf — the main photosynthetic surface. It has an **apex** (tip — described as acute, acuminate, obtuse, etc.), a **margin** (edge — entire, serrate/toothed, dentate, lobed, etc.), and a **base** whose shape (rounded, tapering, heart-shaped, and so on) is also used in leaf identification.



# 3.2 External Structures

- **Venation**: the pattern of veins in the blade. **Midrib** — the stout central vein, a direct continuation of the petiole, from which finer **lateral (secondary) veins** branch off. Two main venation types: **reticulate (net) venation**, in which veins branch repeatedly and rejoin to form a network — typical of dicot leaves; and **parallel venation**, in which the main veins run alongside one another along the length of the blade — typical of monocot leaves.
- **Simple vs. compound leaves**: a **simple leaf** has one undivided lamina; a **compound leaf** has its lamina divided all the way to the midrib into separate leaflets, arranged either pinnately (feather-like, along a central axis) or palmately (radiating from one point, like fingers).



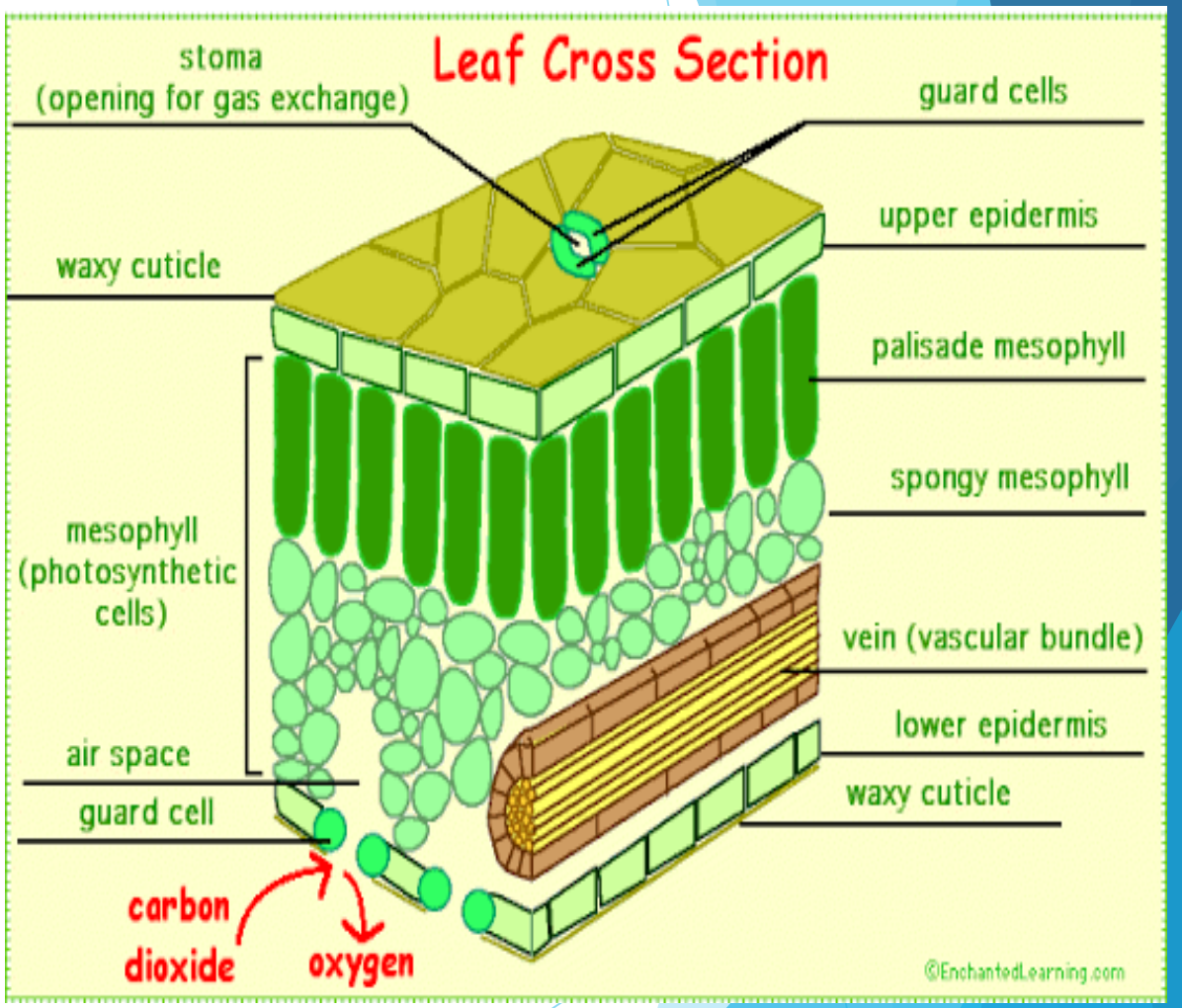
# 3.3 Stomatal Distribution

- A stoma (plural: stomata) is a microscopic pore in the leaf epidermis, bordered by a pair of specialised, bean- or dumbbell-shaped **guard cells**. Unlike ordinary epidermal cells, guard cells contain chloroplasts and can actively change shape to open or close the pore, thereby regulating both gas exchange (CO2 in, O2 out) and water loss through transpiration.
- **Hypostomatic leaves**: stomata are concentrated mainly (or only) on the **lower (abaxial)** surface. This is the commonest pattern in horizontally held dicot leaves, since the shaded, cooler lower surface loses less water through the open pores — see the lower epidermis in Figure 8.
- **Epistomatic leaves**: stomata occur mainly on the **upper (adaxial)** surface — typical of floating aquatic leaves such as the water lily, whose lower surface is in contact with water and would be useless for gas exchange with air.
- **Amphistomatic leaves**: stomata are distributed roughly equally on **both surfaces** — common in many monocots and grasses, whose narrow, often vertically held leaves receive light (and need gas exchange) on both sides.
- **Astomatic leaves**: stomata are absent altogether — typical of fully submerged aquatic leaves, which exchange gases directly across the thin epidermis.
- Stomatal density and distribution are also influenced by light intensity, humidity, $CO_2$ concentration, and the species' native habitat — topics returned to under transpiration in Part B of the course.

# 3.4 Internal Mesophyll Anatomy

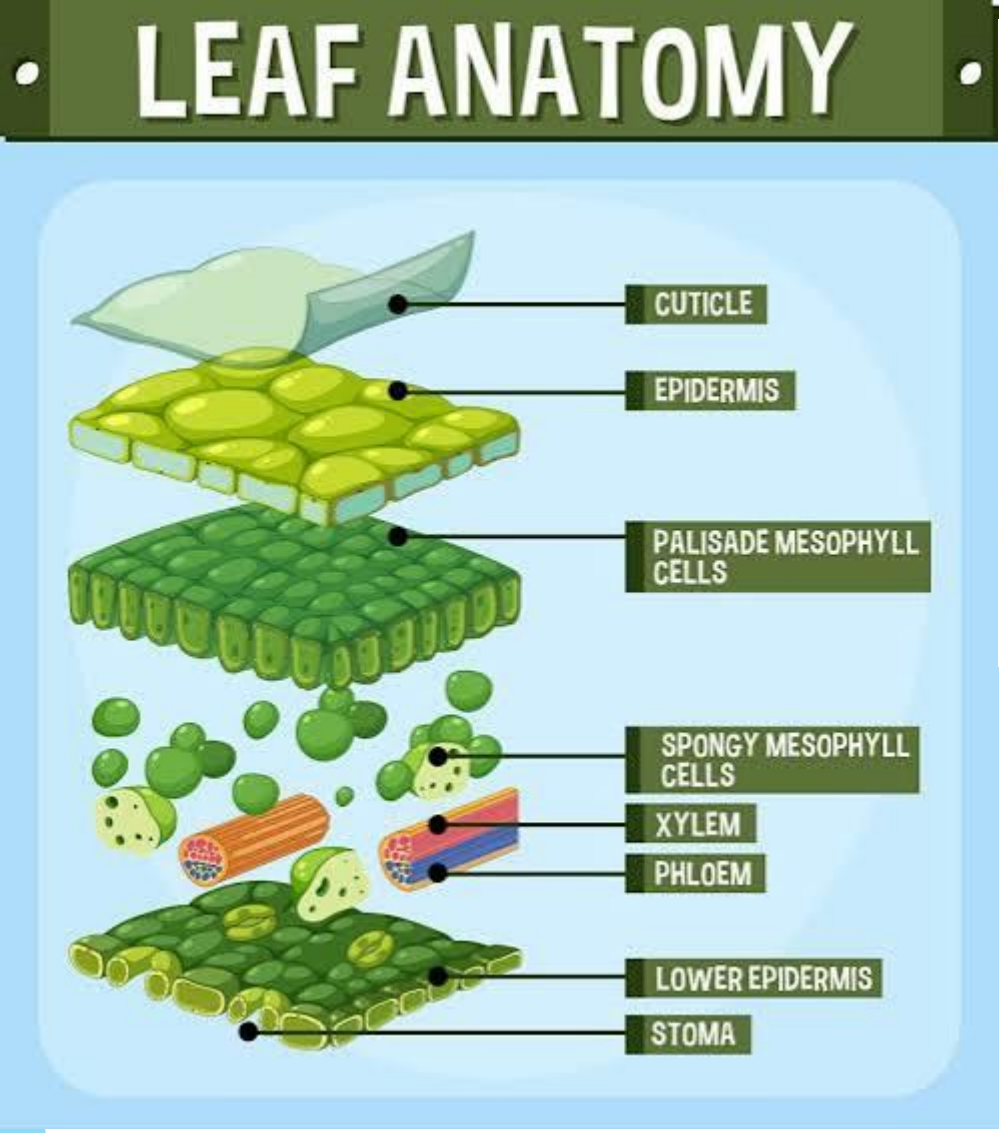
In a typical dorsiventral (bifacial) dicot leaf, a transverse section shows the following layers from top to bottom:

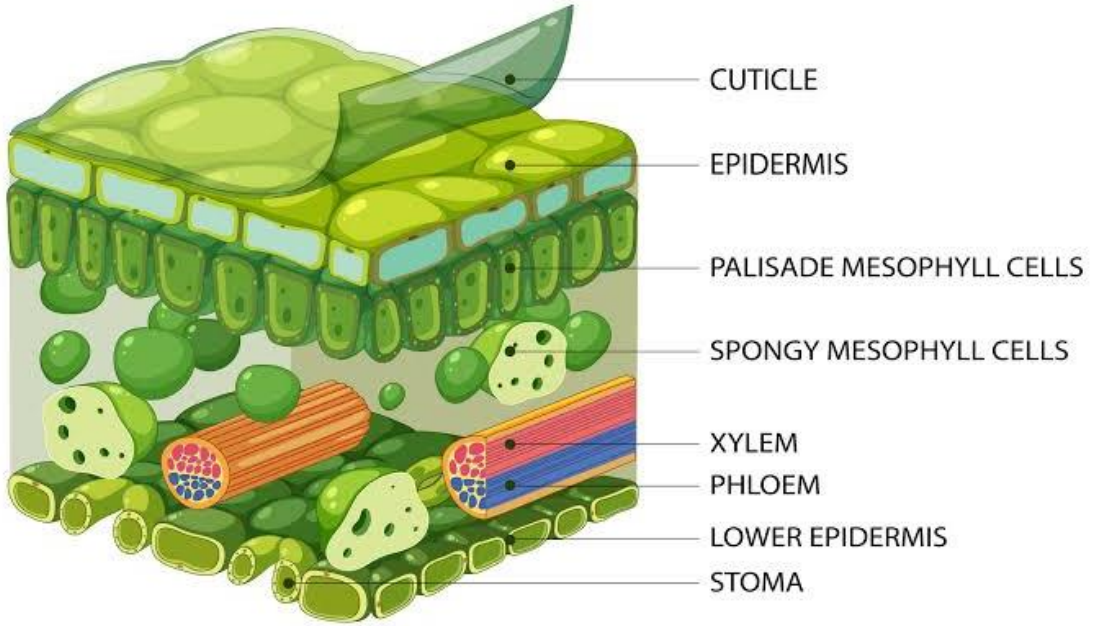
- **Cuticle**: a thin, waxy, waterproof layer covering the upper epidermis (and, more thinly, the lower epidermis); it reduces uncontrolled water loss from the leaf surface.
- **Upper epidermis**: a single, compact layer of transparent, tightly fitted cells (usually without chloroplasts) that lets light pass through to the tissue below, while protecting it.
- **Palisade mesophyll**: lies just beneath the upper epidermis; made up of elongated, columnar cells packed closely together in one or more vertical rows, densely filled with chloroplasts. Its cells and their vertical orientation make it the main site of photosynthesis in the leaf.
- **Spongy mesophyll**: lies below the palisade layer; made up of irregularly shaped, loosely arranged cells with large **intercellular air spaces** between them. These air spaces connect to the stomata and allow $CO_2$ and $O_2$ to diffuse efficiently to and from the photosynthesising cells.

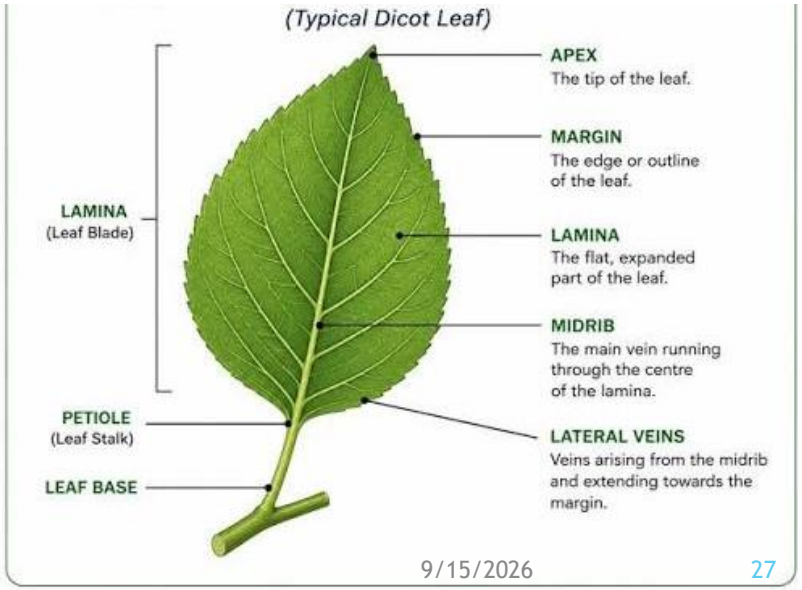


# 3.4 Internal Mesophyll Anatomy

- **Vascular bundles (veins):** embedded within the mesophyll (seen in cross-section as a vein running through the spongy layer); **xylem** is oriented toward the upper (adaxial) side to deliver water to the photosynthetic tissue, while **phloem** is oriented toward the lower (abaxial) side to export the sugars produced; a layer of **bundle sheath** cells surrounds the vein.
- **Lower epidermis:** similar to the upper epidermis but usually bears most of the leaf's **stomata** (in hypostomatic leaves), each flanked by a pair of guard cells, and is covered by a thinner cuticle.
- **Isobilateral leaves** (common in many monocots that hold their leaves vertically) lack this clear palisade/spongy distinction — the mesophyll on both sides of the leaf looks similar, and stomata are usually present on both surfaces (amphistomatic), matching the fact that both surfaces receive comparable light.


LEAF ANATOMY
CUTICLE
EPIDERMIS
PALISADE MESOPHYLL CELLS
SPONGY MESOPHYLL CELLS
XYLEM
PHLOEM
LOWER EPIDERMIS
STOMA


CUTICLE
EPIDERMIS
PALISADE MESOPHYLL CELLS
SPONGY MESOPHYLL CELLS
XYLEM
PHLOEM
LOWER EPIDERMIS
STOMA

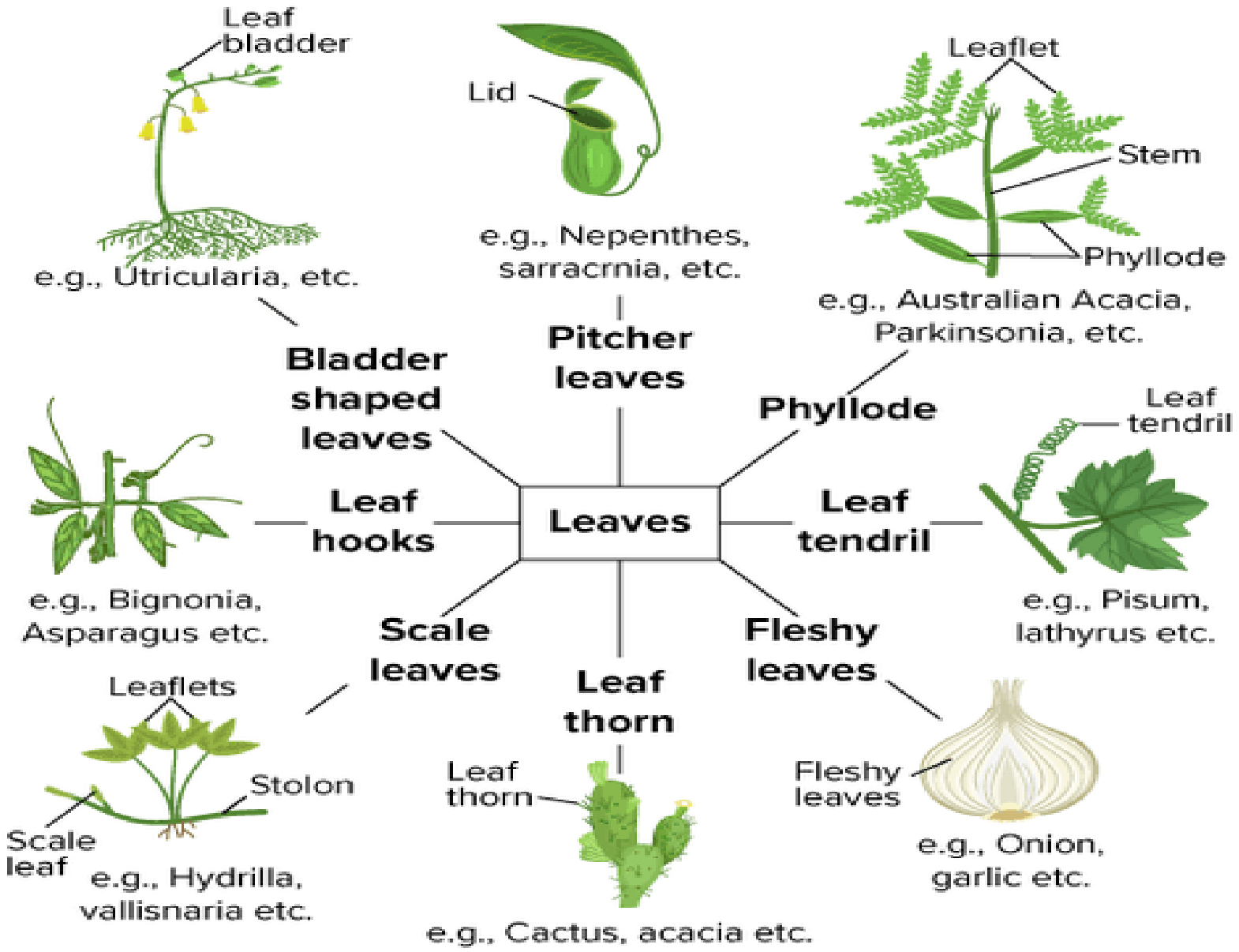

(Typical Dicot Leaf)
APEX
The tip of the leaf.
MARGIN
The edge or outline of the leaf.
LAMINA
The flat, expanded part of the leaf.
MIDRIB
The main vein running through the centre of the lamina.
LATERAL VEINS
Veins arising from the midrib and extending towards the margin.
LAMINA
(Leaf Blade)
PETIOLE
(Leaf Stalk)
LEAF BASE

# 3.5 Leaf Modifications

- **Tendrils**: a leaf, leaflet, or leaf tip modified into a slender, coiling thread used to grip a support while climbing — e.g., the terminal leaflets of the garden pea (Pisum), the leaf tips of Gloriosa, and the tendrils of many cucurbits.
- **Spines**: a leaf, or parts of a leaf, hardened and sharpened, mainly to deter browsing herbivores and to cut down the surface area available for water loss in dry habitats; in such plants (e.g., cactus/Opuntia), photosynthesis is instead carried out by the green, fleshy stem.
- **Storage (succulent) leaves**: thick, fleshy leaves with enlarged, water-storing parenchyma (sometimes called hydrenchyma), adapted to conserve moisture in arid conditions or store food — e.g., Aloe vera, Bryophyllum, and the fleshy scale leaves that make up an onion bulb.

# 3.5 Leaf Modifications

- **Phyllode**: a flattened, leaf-like petiole that takes over photosynthesis while the true blade is reduced or absent — e.g., many Australian Acacia species.
- **Insectivorous (carnivorous) leaves**: modified into traps — a rolled pitcher in Nepenthes, or a small bladder in Utricularia — that supplement the plant's nutrition (especially nitrogen) in nutrient-poor soils.
- **Bud scales**: small, tough, modified leaves that wrap around and protect a dormant bud.
- **Bracts**: reduced leaves, sometimes brightly coloured, associated with flowers or inflorescences — e.g., the colourful bracts of Bougainvillea (the true flowers are small and inconspicuous).


Leaves
Bladder shaped leaves
Leaf bladder
e.g., Utricularia, etc.
Pitcher leaves
Lid
e.g., Nepenthes, sarracrnia, etc.
Phyllode
Leaflet
Stem
Phyllode
e.g., Australian Acacia, Parkinsonia, etc.
Leaf hooks
e.g., Bignonia, Asparagus etc.
Leaf tendril
Leaf tendril
e.g., Pisum, lathyrus etc.
Scale leaves
Leaflets
Stolon
Scale leaf
e.g., Hydrilla, vallisnaria etc.
Leaf thorn
Leaf thorn
e.g., Cactus, acacia etc.
Fleshy leaves
Fleshy leaves
e.g., Onion, garlic etc.

# Quick Revision Summary: Dicot vs. Monocot Vegetative Features

| Feature | Dicot | Monocot |
|---|---|---|
| **Root system** | Taproot (persistent primary root) | Fibrous (adventitious, uniform roots) |
| **Root xylem** | Star-shaped, usually 2–6 arms (di- to hexarch) | Polyarch (many arms), often with central pith |
| **Stem vascular bundles** | Arranged in a ring | Scattered throughout ground tissue |
| **Stem bundle type** | Open (cambium present — secondary growth possible) | Closed (no cambium — no secondary growth) |
| **Cortex / pith in stem** | Clearly differentiated | Not clearly differentiated |
| **Leaf venation** | Reticulate (net-like) | Parallel |
| **Leaf mesophyll** | Usually dorsiventral (palisade + spongy layers) | Often isobilateral (uniform mesophyll) |
| **Stomatal distribution** | Often hypostomatic (mainly lower surface) | Often amphistomatic (both surfaces) |

# Further Reading

- ***Core plant anatomy & morphology***
- Esau, K. (1977). **Anatomy of Seed Plants** (2nd ed.). John Wiley & Sons. — The classic, most detailed reference on plant tissue structure and development; excellent for the internal anatomy sections (§1.2, §2.1, §3.4).
- Pandey, B. P. **Plant Anatomy**. S. Chand & Company. — A widely used undergraduate anatomy text with clear labelled diagrams of root, stem, and leaf transverse sections.
- Pandey, B. P. (2001). **A Textbook of Botany: Angiosperms** (4th ed.). S. Chand & Company. — Covers morphology, anatomy, and economic botany together; useful for linking structure to crop examples.
- Fahn, A. **Plant Anatomy** (4th ed.). Pergamon Press. — A comprehensive anatomy reference with strong comparative coverage of dicot and monocot tissue organization.
- ***General botany / plant biology***
- Evert, R. F., & Eichhorn, S. E. (2013). **Raven Biology of Plants** (8th ed.). W. H. Freeman/Macmillan Learning. — A well-illustrated general botany text covering morphology, anatomy, and diversity.
- Bidlack, J. E., & Jansky, S. H. (2022). **Stern's Introductory Plant Biology** (15th ed.). McGraw-Hill. — An accessible introductory text with dedicated chapters on roots, stems, and leaves.
- ***Plant physiology (for Topic B, and for function behind the structures above)***
- Taiz, L., Møller, I. M., Murphy, A., & Zeiger, E. (2022). **Plant Physiology and Development** (7th ed.). Oxford University Press / Sinauer Associates. — The standard upper-level plant physiology reference; useful once you reach water relations, mineral nutrition, and photosynthesis.
- Hopkins, W. G., & Hüner, N. P. A. **Introduction to Plant Physiology** (4th ed.). John Wiley & Sons. — A more concise, undergraduate-friendly alternative to Taiz & Zeiger.
- *Availability varies by library/edition — any recent edition of these titles will cover the same core content. Where possible, borrow the reference texts (Esau, Fahn) for anatomy detail and a general text (Raven, Stern's) for a first read-through.*